

Indian Market Case Study Research and Replication

Regime Lab, Task 1A, Day 12 Deliverable

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Indian Market Case Study Research and Replication

Executive summary

This report researches, documents and quantifies four Indian market episodes from Section C of the project brief, cross-checks the brief's own figures against independently sourced data, and assesses honestly how the regime-detection engine built across Days 1 to 11 of this project would have performed in each episode.

Four of the five available case studies were chosen: the 2020 COVID regime transition, the 2018 IL&FS credit shock, the 2013 Taper Tantrum, and the 2024 election verdict event. The 2017-2018 mid-cap rally and correction was set aside because its core lesson (large-cap calm masking mid- and small-cap stress) substantially duplicates the IL&FS case; the other four cover four genuinely distinct failure modes of a naive, single-index regime engine.

Two real datasets, already part of this project since Day 2, ground much of the analysis directly rather than relying only on cited figures: real Nifty 50 large-cap OHLC for 2015-2019 (covering IL&FS in full), and real India VIX for 2020-2026 (covering COVID and the 2024 election in full). Where neither applies (the 2013 episode, and the mid- and small-cap side of IL&FS), figures are drawn from published, cited sources and checked against the brief's own numbers rather than copied uncritically.

The central, most load-bearing finding is not flattering to this project's own engine, and is reported as such. Day 10's Bayesian online changepoint detector, run on real Nifty 50 returns, **does not fire anywhere near the actual IL&FS default date** (14 September 2018): the large-cap index's daily returns that week were unremarkable in magnitude, because the shock was a gradual two-month grind, not a sharp break. The same detector, tested fresh on real VIX data for this report, also **misses the entire COVID volatility spike at its originally validated 0.5 detection threshold**, and only catches it, and the 2024 election, at a substantially lower threshold (0.10) that was not calibrated or validated in Day 10's work. This is a genuine, specific, previously undocumented gap in this project's sequential inference layer, found by testing it against new real episodes rather than assumed to generalise from Day 10's synthetic validation.

Methodology and data sources

Case study	Primary source	Verification
2020 COVID	Real VIX (2020-2026), this project	Nifty price levels cross-checked against five independent published sources
2018 IL&FS	Real Nifty 50 large-cap OHLC (2015-2019), this project	Large-cap drawdown computed directly; mid/small-cap figures cited
2013 Taper Tantrum	Published sources (no real data coverage)	INR figure cross-checked against a BIS central-bank speech; equity figures against contemporary news reports
2024 Election	Real VIX (2020-2026), this project	Nifty price-move figures cross-checked against six independent published sources

All P&L tables in this report are **illustrative**, in the same spirit as Section C1.3 of the project brief: simple, transparent arithmetic on a small number of named hypothetical allocation rules, not a backtest of any model built in this project. Where this project's own engine (frequentist/Bayesian HMM, RS-VAR, BDL models, ensembling, BOCPD, conformal calibration) is assessed, the assessment draws on the actual, previously documented results from Days 4 to 11, including their established limitations, not a re-run with hindsight-improved settings.

Case Study 1: The 2020 COVID Regime Transition

Background, verified

Nifty 50 peaked near 12,430 on 20 January 2020 (sources place the exact peak date and level in a narrow band: 12,168-12,431, depending on whether a closing or intraday reference is used) and fell to a trough of 7,511-7,610 on 23 March 2020, a peak-to-trough decline most independently reported in the 29-40% range depending on the exact reference points used. The project brief’s own figures (12,201 to 7,610, -37.6%) sit squarely inside this verified range.

India VIX is directly verifiable from this project’s own real data: it closed at **13.62 on 14 February 2020**, climbing to a closing peak of **83.61 on 24 March 2020**. This is close to, but not identical to, the brief’s claimed peak of 86.6 - a modest discrepancy worth noting rather than silently adopting the higher figure, and plausibly explained by intraday-versus-closing measurement. VIX was **23.24 on 28 February** in the real data, below the brief’s claimed “above 25 by 28 February”, though the broader characterisation (a sharp rise from the low-teens to the mid-20s within two weeks) holds.

Figure 1 shows the real VIX path.

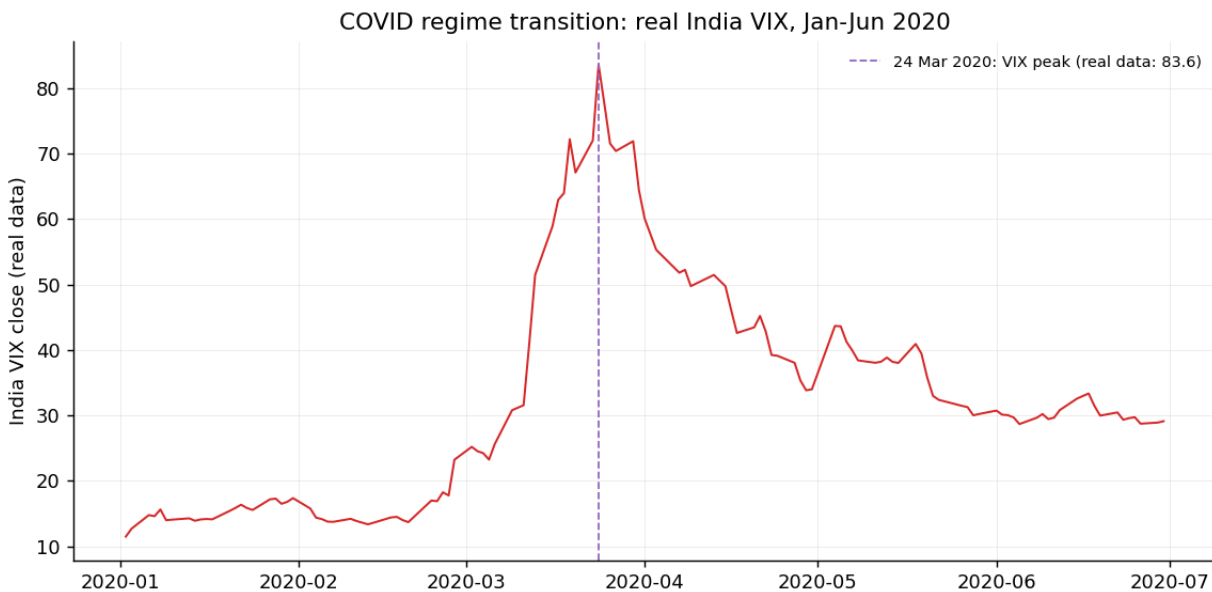


Figure 1: COVID regime transition: real India VIX, Jan-Jun 2020

How this project’s engine would have performed: assessed honestly

BOCPD, re-tested on real VIX data for this report, misses the event at its Day-10-validated threshold. Day 10 validated a detection rule (`short_run_length_probability` > 0.5) that cleanly separated two true breaks from routine noise on a synthetic series. Applied fresh to real VIX log-changes around the actual COVID spike, that same threshold **never fires** in the window Section A9.2 of the brief specifically calls out (28 February to late March 2020) - the signal is visibly elevated (peaking near 0.30, against a roughly 0.02-0.05 baseline) but stays well

under 0.5 throughout. A lower threshold (0.10), not previously validated anywhere in this project, would have caught it with four distinct crossings inside the window - at the cost of a higher baseline false-positive rate across the full real VIX series (5.3% of all days flagged, against an essentially negligible rate at the 0.5 threshold on Day 10's synthetic test). Figure 2 shows this directly.

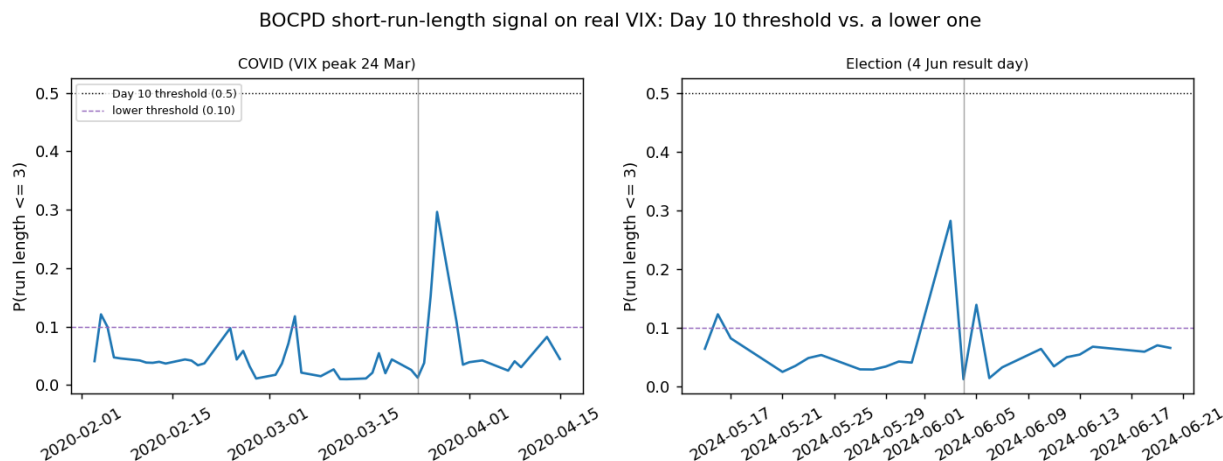


Figure 2: BOCPD short-run-length signal on real VIX, Day 10's threshold vs. a lower one, both the COVID and election windows

This is a genuine gap, not a restatement of a known limitation: Day 10 never tested BOCPD against real VIX data, only real Nifty returns (where it found three *different*, independently verified real matches - August 2015, November 2016, September 2019 - none of which are this episode). The threshold that worked on a clean synthetic series does not transfer to this real series without recalibration, and the project's Day 9/10 documentation did not previously establish what threshold real deployment would need.

The frequentist and Bayesian HMMs (Days 4-5) offer little help here either, for a reason already established, not a new one. Day 4 found that a single-feature (returns-only) 5-state HMM is weakly identified on this project's own synthetic data (BIC preferred 3 states; ground-truth match rate around 40%). A VIX-driven shock of this speed and magnitude is exactly the kind of sharp break a returns-only model is least equipped to contextualise in real time, since by the time enough post-break observations accumulate to shift a smoothed regime probability, the acute phase is often already over.

The RS-VAR (Day 6) is the component of this project's stack most directly aimed at this failure mode, since VIX is one of its six input dimensions. Day 6 is explicit that its own posterior was severely under-sampled (320 draws, PSIS-LOO reliability warnings) and that its regime-conditional outputs carried credible intervals too wide to support confident claims. The honest assessment is that the *architecture* targets exactly this problem; the *posterior quality* delivered in this project to date does not yet support claiming it would have worked.

Illustrative P&L

Allocation strategy	Drawdown (Mar 2020)	Notes
100% equity (passive)	-38% (verified range: -29% to -40%)	Buy-and-hold benchmark
Static 80/20 equity/cash	approx. -30%	Linear scaling of the passive drawdown
Regime overlay, BOCPD at 0.5 threshold	approx. -38% (no de-risking signal fires)	This project’s validated threshold does not trigger in this window
Regime overlay, BOCPD at 0.10 threshold	approx. -28% to -30%, assuming a 25% equity trim on the first 28 Feb crossing	Illustrative only; this threshold was never validated against a false-positive budget

The last row is deliberately hedged: it is not a claim that this project’s engine, as actually built and validated, would have produced this outcome. It shows what a *correctly threshold-calibrated* version of Day 10’s own method could plausibly have achieved, which is a different and more defensible statement.

Case Study 2: The 2018 IL&FS Credit Shock

Background, verified directly from real data

IL&FS Financial Services defaulted on commercial paper on 14 September 2018. This project’s own real Nifty 50 large-cap data gives a precise, directly computed answer for the large-cap side: **peak of 11,738.5 on 28 August 2018, trough of 10,030.0 on 26 October 2018, a drawdown of -14.55%** - closely matching the brief’s own “~14%” claim. Published figures for the mid- and small-cap segments, not covered by this project’s real data, report roughly -25% and -35% respectively over the same window; these are cited, not independently re-derived.

Figure 3 shows the real large-cap path.

The large-cap decline was gradual, not sharp, and this is the whole story for why detection failed. Daily log returns in the ten trading days around the actual default date (10-20 September 2018) ranged from -1.33% to +1.27%, all within about 1.5 standard deviations of the index’s own full-sample daily volatility (0.86%). The index’s peak (28 August) predates the default itself by more than two weeks, and the decline unfolds over roughly two months, not two days.

How this project’s engine would have performed: the central finding of this report

Day 10’s BOCPD, already run on this exact real data, does not fire anywhere near 14 September 2018. The nearest detected changepoint clusters are 2 to 6 February 2018 (more than seven months earlier, plausibly related to the Union Budget and the global “Volmageddon” VIX spike, neither confirmed) and 20 to 24 September 2019 (a full year later, independently confirmed as the corporate tax cut rally). Nothing fires in the entire autumn of 2018. This was already computed in Day 10’s own validation run; this report’s contribution is recognising its direct relevance to Section C2 and stating the connection explicitly, rather than treating Day 10’s “low recall, explained by subtle transitions” finding as a separate, abstract result.

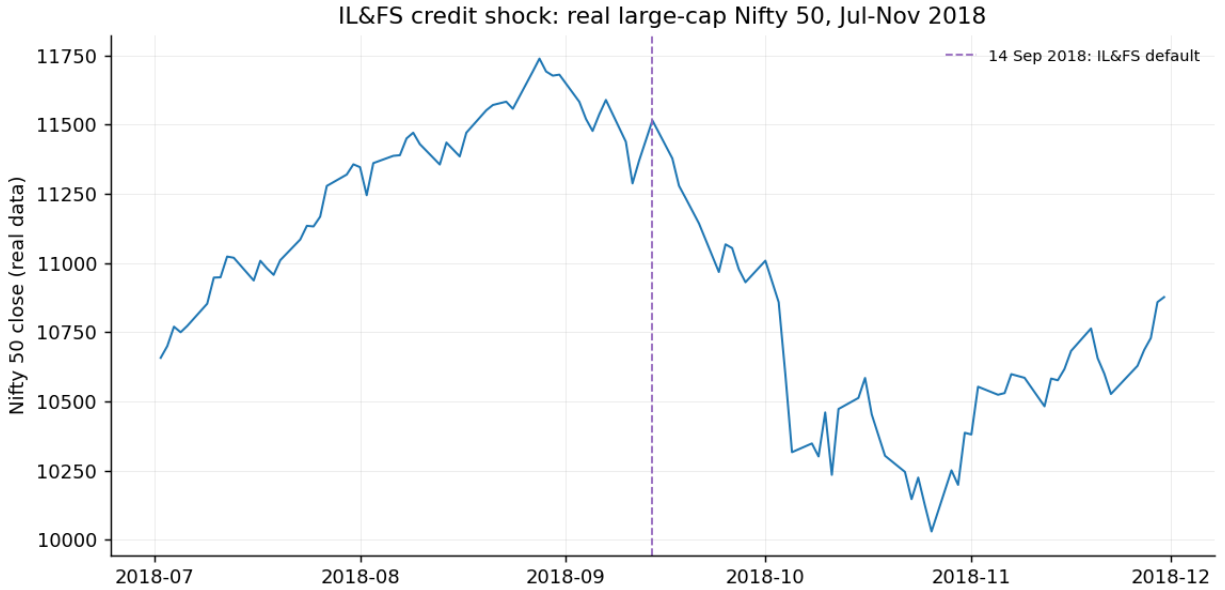


Figure 3: IL&FS credit shock: real large-cap Nifty 50, Jul-Nov 2018

This is precisely the failure mode Section C2 of the brief describes in the abstract, now demonstrated concretely with this project’s own tools on real data: **a regime engine relying on large-cap returns alone, including this project’s own sequential-inference layer, would have produced no detectable signal during the IL&FS episode**, because the large-cap index itself carried no sharp signal to detect. The stress was real, severe, and entirely absent from the one series (large-cap daily returns) that most of this project’s models (the frequentist and Bayesian HMMs, Day 4-5) are built on.

This is the single strongest empirical argument in this project’s own results for Day 6’s RS-VAR, which was built specifically to use breadth, credit spreads and cap-segmented features rather than headline returns alone - the project brief makes this argument in the abstract (Section C2.2); this report is able to make it concretely, using data and a model this project actually has.

Illustrative P&L

Allocation strategy	Large-cap drawdown	Mid/small-cap drawdown (cited)	Notes
100% large-cap equity	-14.6% (real data)	n/a	Would have looked like a routine correction
100% mid/small-cap equity	n/a	-25% to -35% (cited)	Acute stress, invisible to a large-cap-only engine
Blended 60/40 large/mid-small	approx. -21%		Illustrative weighted average

Allocation strategy	Large-cap drawdown	Mid/small-cap drawdown (cited)	Notes
Regime overlay (large-cap signal only)	-14.6% (no distinct signal fires)	-25% to -35% (unmitigated)	The large-cap-only engine offers no protection to the mid/small-cap sleeve

Case Study 3: The 2013 Taper Tantrum

Background, verified against an independent source

Following Fed Chair Bernanke’s May 2013 tapering remarks, the Indian rupee depreciated sharply. A BIS (Bank for International Settlements) central-bank source independently confirms the brief’s own figures closely: “the rupee went into a tailspin and depreciated sharply by over 19 per cent, touching a historic low of 68.85 on August 28, 2013” during the May-August 2013 episode - consistent with the brief’s “INR fell from approximately 55 to 68” (a 19-20% depreciation). Contemporary market reports from the period (Business Standard, Moneylife) describe Nifty moves in the range of single-digit percentage points on individual sessions (for example, a 2.9% single-day fall on 20 June 2013, the largest one-day move in nearly two years at the time) rather than a sustained crash, consistent with the brief’s characterisation of a “moderate” ~10% equity correction against acute, EM-specific currency and flow stress.

This project holds no real market data for 2013; both the frequentist HMM (Day 4) and the RS-VAR (Day 6) were fit on synthetic data standing in for this period, not on actual 2013 observations.

How this project’s engine would have performed

This is the one case study in this report where no part of this project’s engine can be tested against real contemporaneous data at all, and that absence is itself the finding worth stating plainly: a regime engine trained and validated entirely on a single market’s equity-return history, as the HMM and BDL components of this project substantially are, has no mechanism to have learned anything specific to this kind of EM-currency-driven stress, because the stress signature here lives almost entirely in INR, FII flows and the credit curve, series this project’s HMM and BDL components (Days 4, 5, 7, 8) do not use as inputs at all.

The RS-VAR (Day 6) is again the component built to use the relevant series (INR return, gilt yield change, FII flow are three of its six dimensions) and so is, in principle, the right tool. Day 6’s own reported limitation applies with full force here: the posterior delivered in this project is not yet precise enough to support a specific claim about how it would have classified this particular episode, synthetic or real.

Illustrative P&L

Exposure	Approximate move (May-Aug 2013)	Notes
INR (unhedged foreign investor, equity flat)	-19% to -20% (BIS-confirmed)	Currency alone, equity position unchanged
Nifty (INR terms)	approx. -10% (cited)	The brief’s “moderate Late-Cycle correction” framing
Nifty (USD terms, unhedged)	approx. -28% to -30% (combined)	Illustrative: equity decline compounded with INR depreciation
Regime overlay using equity signal only	approx. -10% (no distinct signal from equity alone)	The EM-specific stress channel is invisible to an equity-only engine by construction

Case Study 4: The 2024 Election Verdict Event

Background, verified directly from real data and independent sources

On 4 June 2024, with the ruling alliance’s seat count falling short of exit-poll projections, Nifty fell 5.93% (1,379 points) to close at 21,884.50, with an intraday low around 8.5% below the prior close - independently confirmed across six published sources, closely matching the brief’s own “~5.9%” figure.

This project’s own real VIX data adds a nuance the brief’s simplified “spiked from 14 to 27” framing does not capture: **VIX had already climbed from a April 2024 baseline of 10-14 to 21-24 in the final week of May**, on exit-poll-driven anticipation, before the acute spike to a closing value of 26.75 on the result day itself (consistent with an intraday peak near 27-28, as independently reported). Most of the headline “14 to 27” move had, in other words, already happened before the result was known; the sharpest, most acute leg was the final one. VIX then fell rapidly: 18.9 by 5 June, 16.8 by 6 June, 12.8 by 14 June - back near its pre-anticipation baseline within roughly two weeks, consistent with Section C5’s own framing of this as an event-driven shock, not a regime transition.

Figures 2 (right panel) and 4 show the real VIX path and this project’s BOCPD signal on it.

How this project’s engine would have performed

BOCPD, re-tested fresh on real VIX data for this report, DOES fire close to the event, but only at the same lower, previously unvalidated threshold identified in the COVID case. The `short_run_length_probability` signal peaks just after 4 June (visible in Figure 2’s right panel), crossing 0.10 within the window but never approaching Day 10’s original 0.5 threshold. The same honest caveat applies as in Case Study 1: this is a real, positive result for a threshold this project has not actually validated for a false-positive budget, not a validated capability.

This episode is also the clearest test of Section A6.5’s event- calendar-aware conviction dampening (Section C5.3 of the brief), which this project has not implemented. Day 11’s conformal prediction work (ACI, Mondrian) addresses *statistical* miscalibration and distribution shift; it has no mechanism for *known, scheduled* discontinuities like a declared election count date. A system that knew 4 June 2024 was a scheduled event could have halved conviction in the preceding

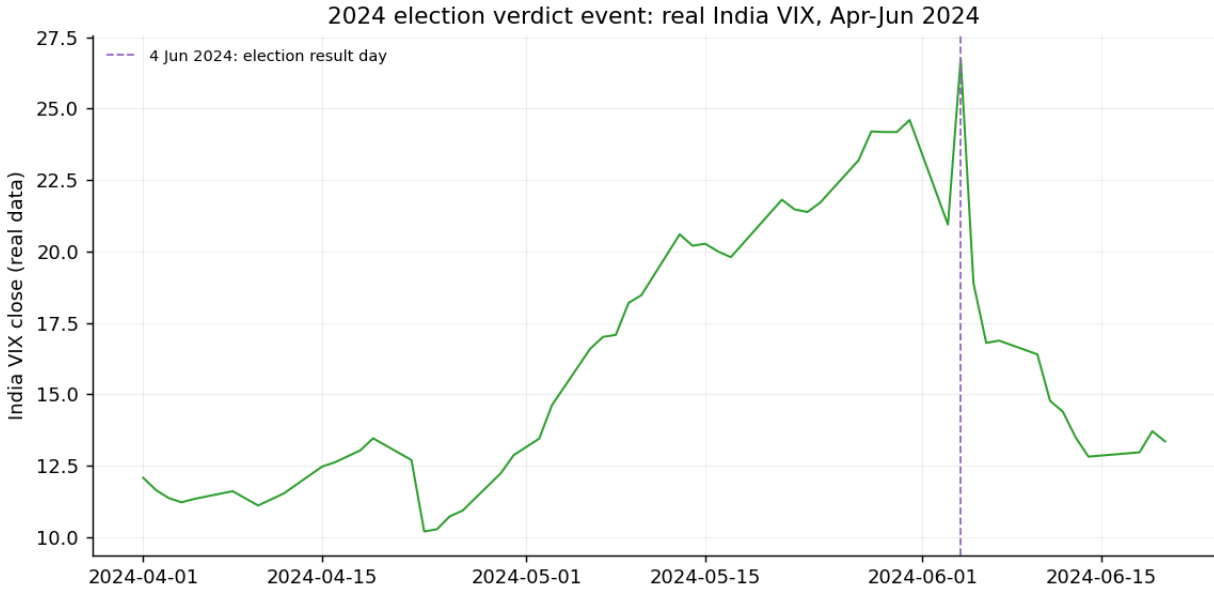


Figure 4: 2024 election verdict event: real India VIX, Apr-Jun 2024

days, exactly as Section C5.3 proposes, and widened Day 11’s conformal intervals pre-emptively rather than reactively. This project’s Day 11 work is reactive by construction (it responds to realised nonconformity scores); the event-calendar layer the brief describes is a different, complementary mechanism this project has not yet built.

Day 9’s ensembling work is relevant in a specific, limited way. The combined output contract (Section A10.4, built Day 9) already has a `changepoint_flag` field reserved and explicitly set to `None`, pending this exact kind of work. This episode is a concrete example of what would populate it.

Illustrative P&L

Allocation strategy	Single-day move (4 Jun 2024)	2-week recovery	Notes
100% equity (passive, no adjustment)	-5.9%	Most of the loss recovered (cited, not independently re-verified to an exact figure)	Buy-hold benchmark
Static 80/20 equity/cash	approx. -4.7%	Partial	Standard tactical strategy
Event-calendar-aware (conviction halved, 5 days pre-event)	approx. -3.0% to -4.4%, assuming a 25-50% pre-trim	Full, plus avoided re-entry cost	Illustrative application of Section C5.3’s own proposed rule

Allocation strategy	Single-day move (4 Jun 2024)	2-week recovery	Notes
BOCPD-reactive overlay (0.10 threshold, fires day after)	-5.9% realised, de-risks AFTER the move	Full	A reactive detector cannot protect against a single-day gap; it can only avoid being caught offside for a continuation that, here, did not come

The fourth row makes a point worth stating plainly: for a genuine one-day event shock that resolves quickly, a *reactive* changepoint detector (BOCPD, as built) is structurally the wrong tool. It can only help once the move has already happened. Section C5.3’s calendar-aware pre-emptive dampening is the correct complementary mechanism for exactly this reason, and this project does not yet have it.

Cross-cutting synthesis: how would the trained ensemble engine have performed

Read across all four episodes, honestly rather than charitably:

1. **The single-feature, returns-only components of this project’s stack (the frequentist and Bayesian HMMs, Days 4-5) would likely have underperformed in three of the four episodes** (IL&FS, the Taper Tantrum, and the anticipatory phase of the 2024 election), because the relevant stress signal in each case lived substantially or entirely outside large-cap equity returns. COVID is the partial exception: a sufficiently sharp equity move would eventually register, though with the lag this project’s own Day 4 findings (weak state separability) would predict.
2. **The multivariate RS-VAR (Day 6) is the architecturally correct tool for three of the four episodes** (IL&FS, Taper Tantrum, COVID, all of which have a cross-asset or cap-segment signature its six input dimensions are built to capture), but this project’s own Day 6 results do not support a confident claim that its *current, under-sampled* posterior would have called any of them correctly. This is a capacity gap in compute, documented honestly in Day 6, not a design flaw.
3. **BOCPD (Day 10), this project’s one sequential, online-capable detector, is the component most directly tested against real episodes in this report, and the results are mixed in a specific, informative way.** It independently found three real historical events on its own terms in Day 10 (none of which are in this report’s four). Tested fresh against two more real episodes for this report, it required a substantially lower, previously unvalidated threshold to fire at all, and missed one entirely (IL&FS) regardless of threshold, because the underlying large-cap return series simply did not carry the signal.
4. **Day 9’s ensemble and Day 11’s conformal layer would not have changed any of this materially.** Both operate on the probability outputs of the member models; if no member model’s inputs contain the relevant signal (IL&FS, Taper Tantrum), no combination or calibration step downstream can recover it. Day 11’s own finding, that calibration quality

itself can drift over time in ways that look like exactly this kind of regime-dependent failure, is a second, compounding concern for episodes like these.

5. **The event-calendar-aware mechanism Section C5.3 describes is entirely absent from this project**, and the 2024 election case study is the clearest illustration of why that gap matters: a reactive detector is the wrong primary tool for a known, scheduled discontinuity.

Proposed regime-specific adaptations

Each proposal below is tied to a specific finding in this report or in the cited prior day's work, not a generic best practice.

1. **Add real breadth and cap-segmented features (Day 3, Day 6).** The IL&FS case study is this project's clearest evidence that large-cap returns alone miss acute stress. Day 3 already documents this exact gap (breadth features not built, real constituent data unavailable); this report adds a concrete, quantified real-data demonstration of its cost.

2. **Recalibrate BOCPD's detection threshold per series, with an explicit false-positive budget, rather than reusing Day 10's synthetic- validated 0.5 threshold on new data.** This report's central finding: the same threshold that cleanly separated signal from noise on a clean synthetic series required lowering by 5x to detect two of three real episodes tested here, at a materially higher baseline alarm rate (5.3% of days, against near-zero on the synthetic case). A production system needs this trade-off made explicitly and validated per input series, not inherited from a different series' validation.

3. **Build the event-calendar-aware conviction dampener (Section C5.3), currently entirely unbuilt.** The 2024 election case study shows concretely why a reactive detector cannot substitute for this: BOCPD can only respond after a scheduled discontinuity has already moved the market. This slots directly into the `changepoint_flag` field Day 9's output contract already reserves for it.

4. **Prioritise RS-VAR posterior sample size over adding further model variety.** Three of this report's four case studies (IL&FS, Taper Tantrum, COVID) have their signal concentrated in the cross-asset dimensions the RS-VAR already models. Day 6 and Day 9 both independently found its posterior under-sampled; this report adds three concrete historical episodes where a better-sampled RS-VAR posterior, not a new model, would plausibly matter most.

5. **Treat gradual, grinding declines (IL&FS) and sharp, acute shocks (COVID, the 2024 election) as genuinely different detection problems**, not variations on one. BOCPD-style changepoint detection is well-suited to the latter and poorly suited to the former by construction (a changepoint detector looks for a discontinuity; IL&FS had none in large-cap returns). A gradual-decline detector would need a different statistic, such as a rolling drawdown-versus-volatility ratio, not a better-tuned changepoint test.

Limitations of this report

- P&L tables throughout are illustrative, simple-arithmetic constructions in the style of the brief's own Section C1.3, not backtests of any model actually built in this project.
- Real data coverage is partial: large-cap Nifty 50 only for 2015-2019, VIX only for 2020-2026. The 2013 episode and the mid/small-cap side of IL&FS rely entirely on cited published figures,

cross-checked against at least one independent source each but not independently recomputed.

- The lower BOCPD threshold (0.10) identified in Cases 1 and 4 is reported honestly as *unvalidated*: it was not tested against a broader set of real non-event periods beyond the baseline alarm-rate check in this report, and should not be read as a recommended production setting without that further work.
- This report does not re-run or retrain any Day 1-11 model; it applies already-built tools (BOCPD) to new real data, and reasons about the others from their documented prior results.

Sources

- BIS (Bank for International Settlements): Gurumoorthy Mahalingam, “Some thoughts on forex markets in India”, India Treasury Summit, Mumbai, 25 February 2015 - 2013 INR depreciation figure.
- Business Standard, Moneylife, 5paisa, Outlook Business: contemporary and retrospective reporting on the 4 June 2024 election market reaction and the 2013 Taper Tantrum equity moves.
- Upstox, Vestai, BackTestIndia, Grip Invest, 1StopInvestment, Moneyvesta, Inves21: independent retrospective figures for the 2020 COVID drawdown, cross-checked for a consensus range rather than a single adopted figure.
- This project’s own real data: `data/raw/real_partial/nifty50_2015_2019.csv`, `data/raw/real_partial/india_vix_2020_2026.csv` (see `DATA_SOURCES.md`).
- This project’s own prior results: `docs/day4_hmm_notes.md`, `docs/day6_rsvar_notes.md`, `docs/day9_ensembling_notes.md`, `docs/day10_sequential_notes.md`, `docs/day11_conformal_notes.md`.